

Tuition Fees Project: Model Skeleton and UK Policy Inputs

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1 Purpose

This note converts the current project scaffold into a compact equation skeleton and records verified institutional inputs for England student finance policy.

2 Model in Words First

2.1 What problem this model is trying to solve

The model asks a narrow but important incidence question: if the same higher-education spending is financed in three different ways, who pays in present-value terms, and at what point in the life cycle? The three funding systems are:

- an income-contingent tuition loan system,
- a graduate tax, and
- financing from broad general taxation.

The objective is not to model university demand or macro feedback in v0. The objective is to map policy rules into distributional incidence transparently.

2.2 Why this is a live policy issue

Policy debate often compares these systems using annual cash flows or average repayments. That can blur the key economic margin: timing. A payment made at age 19 is not equivalent to the same nominal payment made at age 40. The model makes this timing margin explicit by converting each regime into discounted lifetime burdens.

There is also a composition issue. Under loan systems, some higher-wealth families may pay fees upfront. If that group exits the borrowing pool, measured subsidy and cross-subsidization in the remaining loan book can shift. The model includes this margin directly.

2.3 What this model can and cannot answer in v0

Can answer:

- How burden differs between graduates and non-graduates across regimes.
- How burden differs across earnings profiles among graduates.
- How an upfront-fee margin by wealth type changes measured incidence.

Cannot answer yet:

- Schooling participation responses to policy (education choice is exogenous).
- Wage-equilibrium or macro feedback effects.
- Behavioral responses in labor supply or avoidance.

2.4 Intuition

The baseline intuition is straightforward:

- Income-contingent loans insure low-earnings graduates because payments depend on realized earnings and unpaid balances can be written off. That tends to compress burdens at the lower end of the graduate earnings distribution.
- Graduate taxes spread cost over graduate income without personal debt balances. This can look similar to income-contingent repayment for some earnings paths, but differs through duration and rate design.
- General taxation broadens incidence to non-graduates by construction, so it can reduce average graduate burden but increases burden outside the graduate group.
- Even when undiscounted totals are similar, PDV burdens can differ materially because the payment timing differs across regimes.

3 Equation Skeleton (v0)

3.1 Indices and primitives

Individual type is $i = (e_i, a_i, g_i)$ where $e_i \in \{G, N\}$ is graduate status, $a_i \in \{H, L\}$ is family wealth type, and g_i indexes an exogenous earnings path $\{y_{i,t}\}_{t=0}^T$. Tuition is F , discount factor is $\beta \in (0, 1)$, and loan interest rate in regime A is r_L .

3.2 Common incidence object

Define net period public payment under regime r as

$$p_{i,t}^r = \text{Payment}_{i,t}^r - \text{Transfer}_{i,t}^r,$$

and PDV burden as

$$B_i^r = \sum_{t=0}^T \beta^t p_{i,t}^r.$$

Pairwise comparison:

$$\Delta B_i^{r,s} = B_i^r - B_i^s.$$

$\Delta B_i^{r,s} < 0$ implies type i prefers regime r to s on PDV burden.

3.3 Regime A: income-contingent tuition loan

Loan participation indicator among graduates: $I_i^{\text{loan}} \in \{0, 1\}$.

$$D_{i,0} = I_i^{\text{loan}} F.$$

Repayment rule:

$$R_{i,t}^A = \phi \max(y_{i,t} - \bar{y}, 0) \mathbf{1}\{D_{i,t} > 0\}.$$

Debt evolution:

$$D_{i,t+1} = \max\{(1 + r_L)D_{i,t} - R_{i,t}^A, 0\}, \quad t = 0, \dots, T-1.$$

Write-off at T_w :

$$D_{i,T_w+1} = 0, \quad R_{i,t}^A = 0 \text{ for } t > T_w.$$

Upfront payment component (if no loan):

$$P_{i,0}^{A,\text{upfront}} = (1 - I_i^{\text{loan}}) \mathbf{1}\{e_i = G\} F.$$

Working-life repayment component:

$$P_{i,t}^{A,\text{repay}} = R_{i,t}^A.$$

Total payment flow:

$$\text{Payment}_{i,t}^A = P_{i,t}^{A,\text{upfront}} + P_{i,t}^{A,\text{repay}}.$$

Baseline incidence accounting sets $\text{Transfer}_{i,t}^A = 0$ unless explicit rebate channels are added.

3.4 Regime B: graduate tax

No personal debt stock. Baseline tax function:

$$\text{Payment}_{i,t}^B = \tau_G y_{i,t} \mathbf{1}\{e_i = G\} \mathbf{1}\{t \leq T_G\}.$$

Optional threshold variant:

$$\text{Payment}_{i,t}^B = \tau_G \max(y_{i,t} - \bar{y}^G, 0) \mathbf{1}\{e_i = G\} \mathbf{1}\{t \leq T_G\}.$$

3.5 Regime C: general taxation finance

No tuition debt and no graduate-specific surcharge.

$$\text{Payment}_{i,t}^C = \tau_{\text{all}} y_{i,t}.$$

3.6 Cohort bookkeeping identities

Higher-education outlay for cohort k :

$$HE_k = F N_{G,k}.$$

Regime A receipts and subsidy:

$$\text{Rev}_k^A = \sum_i \sum_t \beta^t R_{i,t}^A, \quad \text{Sub}_k^A = HE_k - \text{Rev}_k^A.$$

Regime B and C receipts:

$$\text{Rev}_k^B = \sum_i \sum_t \beta^t \text{Payment}_{i,t}^B, \quad \text{Rev}_k^C = \sum_i \sum_t \beta^t \text{Payment}_{i,t}^C.$$

4 UK Policy Inputs (England, verified on 2026-02-23)

4.1 Tuition fee caps

For 2025/26 (from 1 August 2025), government-published maximum fees include:

- GBP 9,535 (standard full-time)
- GBP 11,440 (accelerated full-time)
- GBP 7,145 (part-time)

[INFERENCE] A baseline calibration value of annual tuition $F = 9,535$ is policy-consistent for a standard full-time course in 2025/26.

4.2 Repayment thresholds and rates

Current thresholds in repayment guidance:

- Plan 2: GBP 28,470
- Plan 5: GBP 25,000

Both plans repay 9% of earnings above the relevant threshold.

4.3 Plan definitions and write-off

- Plan 2: undergraduate starters from 1 September 2012 to 31 July 2023; balance cancellation after 30 years.
- Plan 5: starters on/after 1 August 2023; balance cancellation after 40 years.

4.4 Interest rules

- Plan 2: RPI-based, income-contingent post-study rates; while studying, normally $\text{RPI} + 3\%$.
- Plan 5: normally RPI only.

4.5 Evidence on upfront payment vs borrowing

SLC reports 7.3% of full-time loan borrowers in 2024/25 took only a Tuition Fee Loan and no Maintenance Loan. SLC also states that estimated tuition-fee-loan take-up against the eligible population is no longer published after a November 2025 method change.

[INFERENCE] Direct measurement of “paid tuition upfront with no tuition loan” is not identified in this published series; scenario shares for upfront payment should be treated as assumptions and tested in sensitivity analysis.

5 Sources

1. GOV.UK, “Changes to tuition fees: 2025 to 2026 academic year” (updated 26 November 2025): [source link](#)
2. GOV.UK, “Repaying your student loan: How much you repay” (accessed 2026-02-23): [source link](#)
3. GOV.UK, “Student loans: a guide to terms and conditions 2025 to 2026” (accessed 2026-02-23): [source link](#)
4. GOV.UK/SLC, “Student support for higher education in England 2025” (updated 11 December 2025): [source link](#)
5. GOV.UK/SLC, “Income Contingent Student Loan repayment plans, interest rates and calculations (England)” (2024/25): [source link](#)